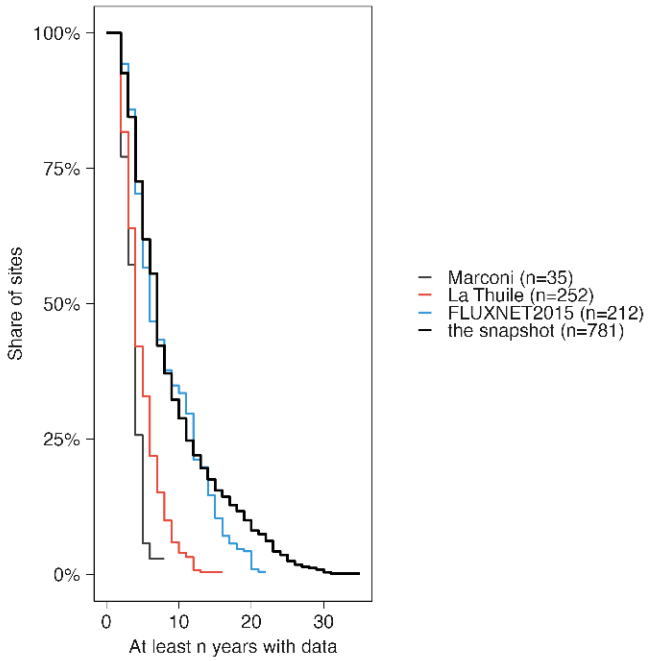
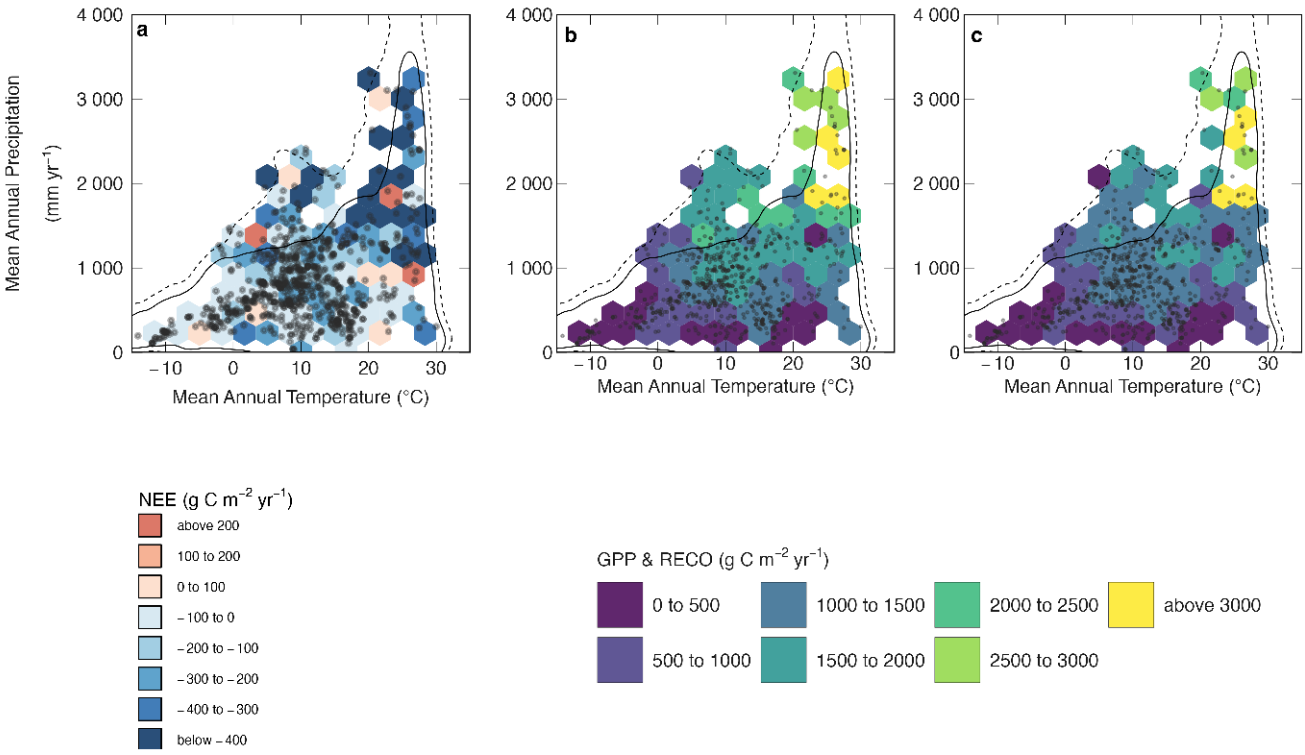
Supplementary Figure S1

For each of four FLUXNET network generations (Marconi, La Thuile, FLUXNET2015, the snapshot), the share of that collection's sites with at least n years of data, as a step line. Data used in the analyses are described in Section 2.5.



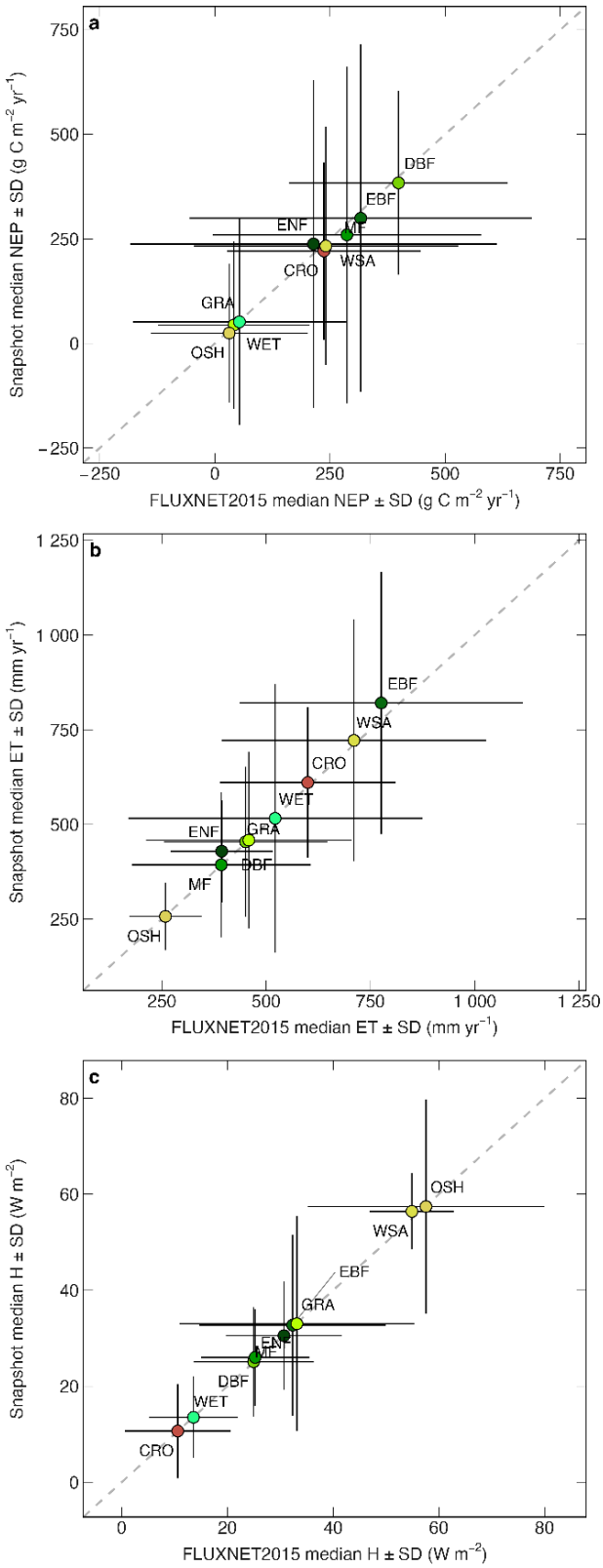
Supplementary Figure S2

Whittaker climate−space distribution of the snapshot network's net ecosystem exchange (NEE), gross primary productivity (GPP) and ecosystem respiration (RECO). Towers are placed in mean−annual−temperature by mean−annual−precipitation space (WorldClim v2.1) and coloured by the hexagon−binned median of each site's flux. Solid and dashed black lines show the 95% and 99% highest−density regions of global ice−free land in the same climate space, for comparison with the main text's equivalent NEE distribution. Data used in the analyses are described in Section 2.5.



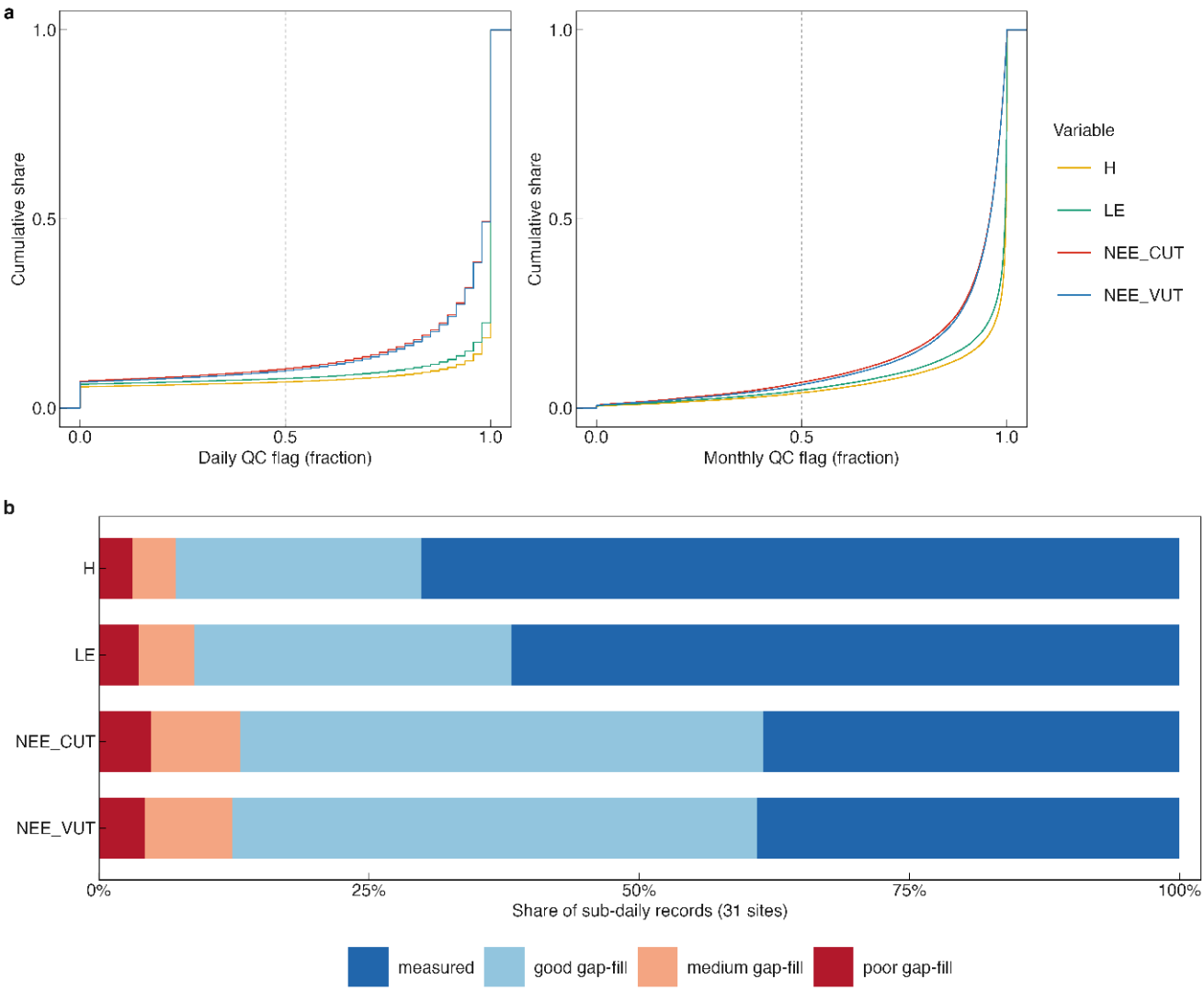
Supplementary Figure S3

Comparison of per−IGBP−class median net ecosystem production (NEP), evapotranspiration (ET) and sensible heat flux (H) between the FLUXNET2015 release and the snapshot, restricted to site−years present and qualifying in both datasets for a given site and calendar year. This matched comparison isolates differences arising from updated flux−processing software from differences arising from network composition change, complementing the full−network comparison in the main text. Data used in the analyses are described in Section 2.6.



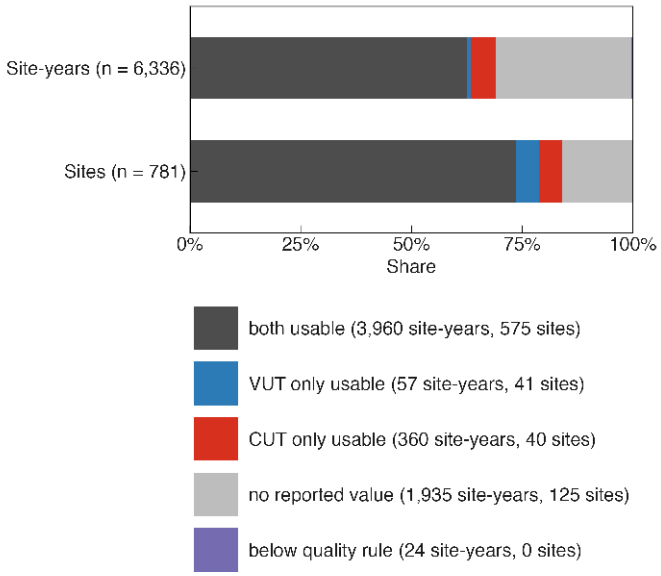
Supplementary Figure S5

Quality−control (QC) flag distributions for four flux variables, and the measured/gap−filled share of sub−daily records at the sites with sub−daily data extracted.



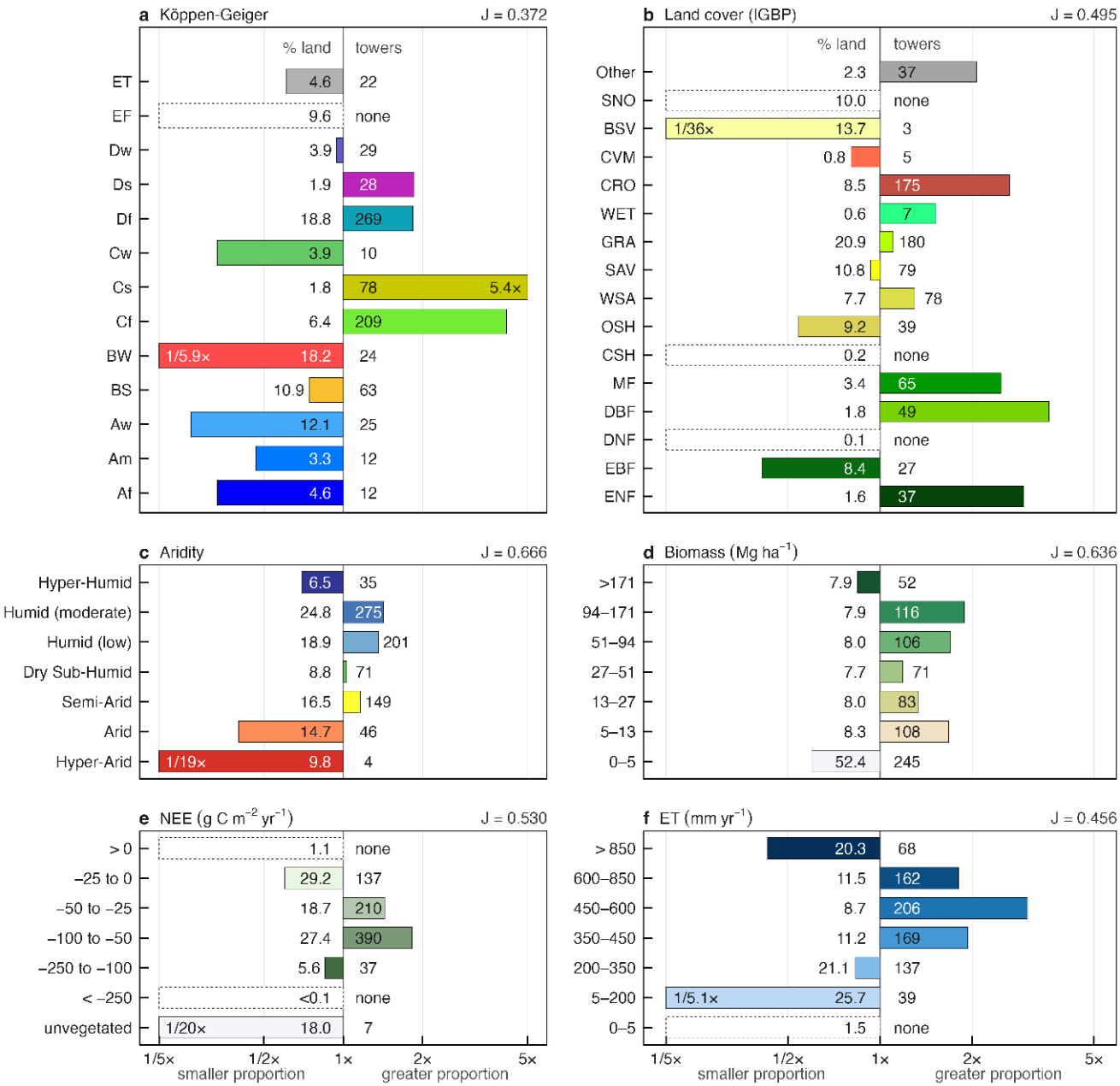
Supplementary Figure S6

Availability of annual NEE across the snapshot, by usability category.



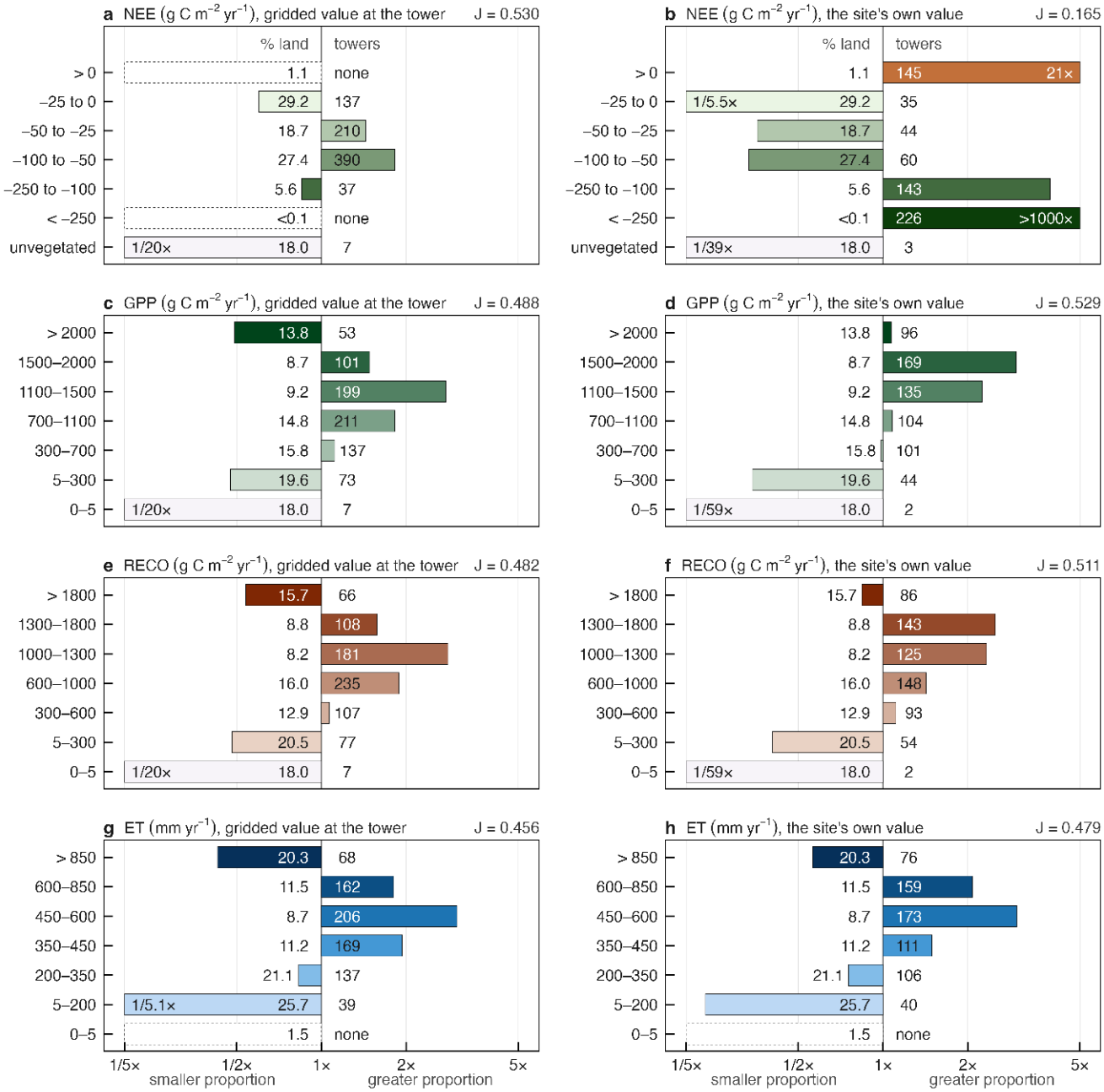
Supplementary Figure S7

Network sampling of the snapshot (n=781 sites) across six axes −− Koppen−Geiger climate class, land cover (IGBP), aridity, aboveground biomass, net ecosystem exchange and evapotranspiration −− using the gridded value at the tower coordinate rather than each site's own measurement. This is the companion to the main text's sampling figure, which instead uses the site's own value; the two together show how much the network's apparent sampling depends on that choice. Network sampling analysis is described in Section 2.7.



Supplementary Figure S8

Network sampling of the snapshot for four fluxes −− net ecosystem exchange, gross primary productivity, ecosystem respiration and evapotranspiration −− each compared using both the gridded value at the tower and the site's own value. Extends the main text's sampling figure (which covers net ecosystem exchange and evapotranspiration) to add gross primary productivity and ecosystem respiration. Network sampling analysis is described in Section 2.7.



Supplementary Figure S9

Network sampling (gridded value at the tower) tracked across four generations of FLUXNET data collection −− Marconi, La Thuile, FLUXNET2015 and the snapshot −− for the same six axes as the main text's sampling figure. Shows how the network's coverage of climate, land−cover, aridity, biomass and flux space has changed as the network has grown. Network sampling analysis is described in Section 2.7.

